

PHASE 1

Ideation Phase

Empathizing with stakeholders and framing the problem PulseView sets out to solve

PulseView – Heart Disease Risk Analysis Dashboard

Project	PulseView – Heart Disease Risk Analysis Dashboard
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1. Introduction

Heart disease remains one of the leading causes of death worldwide, and its impact keeps growing alongside modern lifestyle changes such as poor diet, inactivity, and rising stress levels. While prevention and early detection make the biggest difference to outcomes, spotting the right patterns in patient demographics, medical history, and lifestyle habits requires the right analytical tools. This document captures the ideation activities carried out before development began — understanding the people affected, framing the problem in their words, and converging on the concept for the **PulseView** dashboard.

2. Problem Statement

Element	Description
Who is affected?	Clinicians, public-health policymakers, and individuals with a personal or family history of heart disease.
What is the problem?	Heart-disease risk factors are scattered across demographic, lifestyle, and clinical records, making it hard to see which factors matter most or how they combine.
Where does it occur?	In hospitals, public-health departments, and at home, wherever someone needs to interpret raw health-survey data quickly.
Why does it matter?	Without a clear, visual read of the risk factors, prevention campaigns and personal lifestyle changes are based on guesswork rather than evidence.

Problem Statement (PSN): “How might we help clinicians, policymakers, and individuals quickly identify the demographic, lifestyle, and clinical factors most strongly linked to heart disease, in a format that needs no data-analysis background to understand?”

3. Empathy Mapping

Three representative personas were used to keep the design grounded in real needs throughout the project, rather than building a generic analytics dashboard.

Persona	Says	Thinks	Does	Feels
Dr. Sharma — Cardiologist	“Which lifestyle factors are really driving heart disease in my middle-aged patients?”	Wants evidence to justify targeted campaigns rather than generic advice.	Segments patients by age, BMI, cholesterol, and smoking status.	Frustrated by scattered records; relieved when patterns are visual and immediate.
Ramesh — Health Policy Officer	“Where should limited public-health budget go first?”	Needs comparisons across rural vs. urban populations to justify policy.	Cross-references sedentary-lifestyle indicators against disease rates.	Under pressure to show measurable impact from prevention programmes.

Persona	Says	Thinks	Does	Feels
Anita, 45 — Patient	“Am I at risk, given my family history?”	Wants a simple, non-technical way to compare herself to healthy benchmarks.	Checks her cholesterol, blood pressure, and habits against the dashboard.	Anxious about family history; wants clear, actionable next steps.

4. Brainstorming & Idea Prioritisation

A short brainstorming pass generated several candidate directions for presenting the dataset. Each was scored informally against three criteria: **clarity** for a non-technical audience, **feasibility** with free/public tooling, and **shareability** outside a desktop analytics tool.

Idea	Clarity	Feasibility	Shareable	Decision
Raw statistical report (PDF/CSV export)	Low	High	Low	Rejected
Standalone Tableau workbook shared as a file	Medium	High	Low	Rejected
Tableau dashboard + guided story, embedded in a lightweight web app	High	High	High	Selected
Full custom web app with a live database backend	High	Low (time-boxed project)	High	Deferred

5. Proposed Solution

The selected concept, named **PulseView**, pairs a Tableau dashboard and a four-scene guided story built from the heart-disease dataset with a lightweight Flask front end. The Flask layer does not reprocess the data itself — it embeds the published Tableau Public views so visitors get an interactive, browser-based experience without installing Tableau, while the site itself supplies context: a home page, an about page explaining the personas and methodology, and a contact page for follow-up questions.

- **Dashboard view:** four cross-tabulated panels — gender vs. heart disease, diabetic vs. stroke history, race-wise prevalence, and smoking/alcohol impact.
- **Story view:** a four-scene guided narrative (Overview → Gender Risk → Lifestyle Factors → Race & Conditions) that walks a first-time visitor through the findings in order.
- **Supporting pages:** Home, About (problem framing + personas + build process), and Contact, so the dashboard sits inside a complete, presentable web experience.

6. Ideation Summary

The ideation phase converged on a solution that is visual-first, requires no login or installation, and speaks to three distinct audiences through one shared dataset. This concept carries forward into the Requirement Analysis phase, where it is broken down into concrete functional and non-functional requirements.